RISHI THAKUR

SOFTWARE DEVELOPMENT ENGINEER

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PROFESSIONAL SUMMARY

Software Development Engineer with a strong computer science foundation (M.Tech CSE, GATE CS qualified) and experience designing backend services, automated data pipelines, and relational databases. Proficient in data structures, algorithms, OOP in Python, and SQL optimization. Experienced in Agile teams across Linux (Ubuntu, Fedora daily driver), Docker containerization, and edge deployment models.

SKILLS

- **Languages & Software Engineering:** Python (OOP, Data Structures), RESTful APIs (FastAPI, Flask), Core Algorithms, PyTest, Unit Testing, Code Refactoring, Clean Architecture
- **Systems, OS & Cloud Platforms:** Linux Administration (Daily Fedora, Ubuntu, Bash), Docker Containerization, Kubernetes Concepts, Virtualization, Edge & Cloud Deployment, CI/CD, Git/GitHub
- **Databases & Data Infrastructure:** Advanced SQL (MySQL, MariaDB, OracleDB) — Joins, CTEs, Window Functions, Schema Design, Indexing, Query Optimization, ETL Pipelines
- **Communication & Spoken Skills:** Agile Sprint Ceremonies, Technical Documentation & Procedures, Cross-Functional Engineering Collaboration, Requirements Elicitation, Code Reviews

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
Software Engineering & Analytics Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Engineered automated backend **Python/ETL pipelines** integrating biometric attendance APIs (AEBAS) with State Treasury databases across **33 districts**.
- Architected and optimized complex **SQL queries and joins** mapping employee records to treasury codes with **100% data fidelity**.
- Constructed a modular **3-tier risk classification service** in Python, enforcing business logic to identify attendance compliance gaps across facilities.
- Reconciled disparate backend database schemas across MariaDB, State Treasury, and modular Next-Gen e-Hospital systems, eliminating data silos.
- Collaborated within an **Agile engineering workflow** with National Informatics Centre (NIC) teams, authoring technical documentation and procedures.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Edge AI Architecture & Telemetry Pipeline*
Technologies: Python, PyTorch, Diffusion Models (DDPM/DDIM), GRU, Scikit-learn, Random Forest, Linux/Bash, Edge Deployment

- Architected a high-throughput Python service pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** simulating signal trajectories from multivariate **spatio-temporal** telemetry.
- Addressed extreme **98%/2% class imbalance** using SMOTE + Tomek Links, training a Random Forest policy achieving a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Scalable Machine Learning & NLP Service*
Technologies: Python, PyTorch, Transformers, BERT, XGBoost, Bi-LSTM, 1D-CNN, Scikit-learn

- Engineered modular NLP classification services: extracted 768-dim dense **BERT embeddings** with **XGBoost**, and built a deep hybrid Bi-LSTM/Transformer model.
- Applied stratified 80/20 sampling, text normalization, and early stopping, achieving **99% Precision** and **98% Recall** in synthetic LLM classification.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **SQL & Database Design** (Udemy)
- **Celonis Process Mining Foundations** (Credly)
- **Microsoft Power BI** (Udemy)
- **IELTS Academic:** 7.5 Band Score